
MUHAMMAD TAHA
22F-3277
COAL LAB 10
27/10/2023

LAB 10

(Total Marks 15)

Task 1: Write a program that print asterisk on first half screen using following instructions:

- Use STOS Instruction
- Set direction flag 1
- Also Use stack and subroutine concepts

org 0x0100

jmp start

clrscr:

```
    push es
    push ax
    push cx
    push di

    mov ax, 0xb800
    mov es, ax
    xor di, di

    mov ax, 0x0720
    mov cx, 2000
    cld
    rep stosw

    pop di
    pop cx
    pop ax
    pop es
    ret
```

print:

```
    push es
    push ax
    push cx
    push di

    mov ax, 0xb800
    mov es, ax
    xor di, di
```

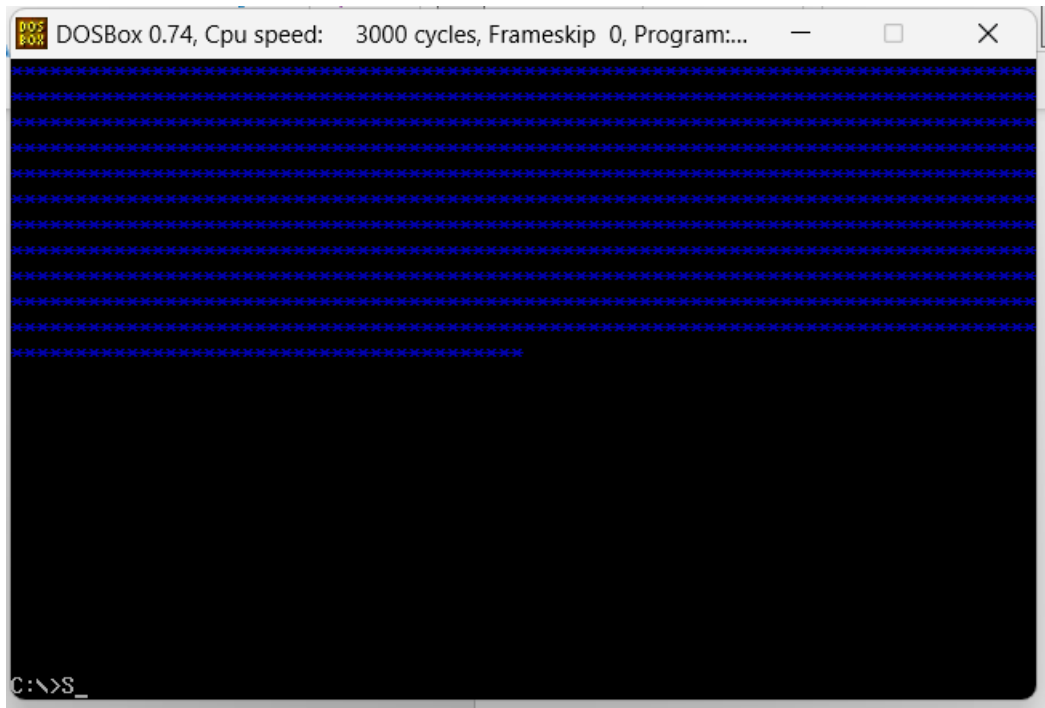
```
mov ax, 0x012A
mov cx, 1000
cld
rep stosw
```

```
pop di
pop cx
pop ax
pop es
ret
```

start:

```
call clrscr
call print
```

```
mov ax, 0x4c00
int 0x21
```



Task 2: Write a program that print the given input string (your full name) in reverse order on the screen with following instructions:

- Using STOS and LODS instruction
- You should clear the screen first and then proceed to print it
- For example: Message: db '**hello world**'
- Output of above string is '**dlrow olleh**'

```
org 0x0100
```

```
jmp start
```

```
message: db 'Muhammad Taha'
```

```
len: dw 13
```

```
clrscr:
```

```
    push es
    push ax
    push cx
    push di
```

```
    mov ax,0xb800
    mov es,ax
    mov cx,2000
    mov ax,0x0720
```

```
    cld
    rep stosw
```

```
    pop di
    pop cx
    pop ax
    pop es
    ret
```

```
print:
```

```
    push bp
    mov bp, sp
    push es
    push ax
    push cx
    push si
    push di
```

```
    mov ax, 0xb800
    mov es, ax
    mov di, 160
```

```
    mov si, [bp+6]    ;point si to string
    mov cx, [bp+4]    ;load string length in cx
    add si, cx
```

dec si

mov ah, 1

next:

std

lodsb

cld

stosw

loop next

pop di

pop si

pop cx

pop ax

pop es

pop bp

ret 4

start:

call clrscr

mov ax, message

push ax

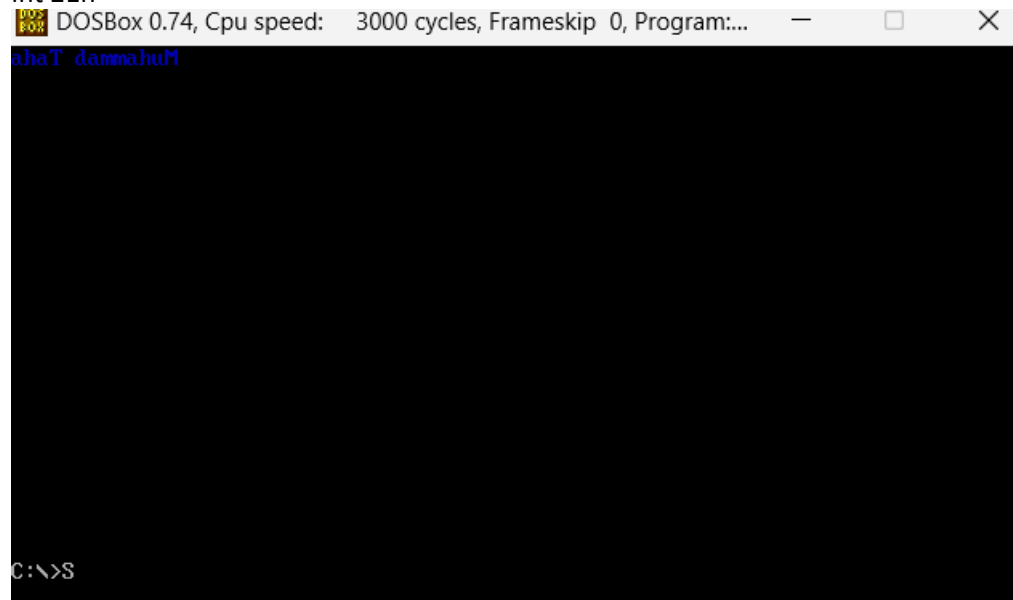
mov ax, [len]

push ax

call print

mov ax, 0x4c00

int 21h



Task3: Write a program that count the number of non-vowels letter in a given string (your full name) and print its vowels letter on the screen

- Using Stos and Lods instruction
- You should clear the screen first and then proceed to print it
- Store the counter in variable
- Like Message: db. '**Assembly Language**'
- Output of above string is '**ssmblylngg**' & counter:10.

```
org 0x0100
```

```
jmp start
```

```
message: db 'Muhammad Taha'
```

```
counter: db 0
```

```
clrscr:
```

```
    push es
```

```
    push ax
```

```
    push cx
```

```
    push di
```

```
    mov ax,0xb800
```

```
    mov es,ax
```

```
    mov cx,2000
```

```
    mov ax,0x0720
```

```
    cld
```

```
    rep stosw
```

```
    pop di
```

```
    pop cx
```

```
    pop ax
```

```
    pop es
```

```
    ret
```

```
print:
```

```
    push bp
```

```
    mov bp, sp
```

```
    push es
```

```
    push ax
```

```
    push bx
```

```
    push cx
```

```
    push si
```

```
    push di
```

```
    mov si, [bp+4]
```

```
    mov cx, [bp+6]
```

```
    mov ax, 0xb800
```

```
    mov es, ax
```

```
    mov di, 160
```

```
    mov ah, 1
```

```

        cld
next:    lodsb

checkvowel:
        cmp al, 'a'
        je skip
        cmp al, 'e'
        je skip
        cmp al, 'i'
        je skip
        cmp al, 'o'
        je skip
        cmp al, 'u'
        je skip
        cmp al, 'A'
        je skip
        cmp al, 'E'
        je skip
        cmp al, 'I'
        je skip
        cmp al, 'O'
        je skip
        cmp al, 'U'
        je skip

        stosw ;prints the non vowel
        cmp al, ' '
        je skip ;skips counting of space
        inc bx ;counts non vowels

skip:
        loop next

        mov [counter], bx ;stores counter in variable

        call printCount

        pop di
        pop si
        pop cx
        pop bx
        pop ax
        pop es
        pop bp
        ret 4

printCount:
        push bx
        push ax
        push di
        push es

        add bl, 0x30
        mov bh, 1

```

```
mov ax,bx
add di, 2
mov [es:di], ax ; print char on screen
```

```
pop di
pop di
pop ax
pop bx
ret
```

start:

```
call clrscr
```

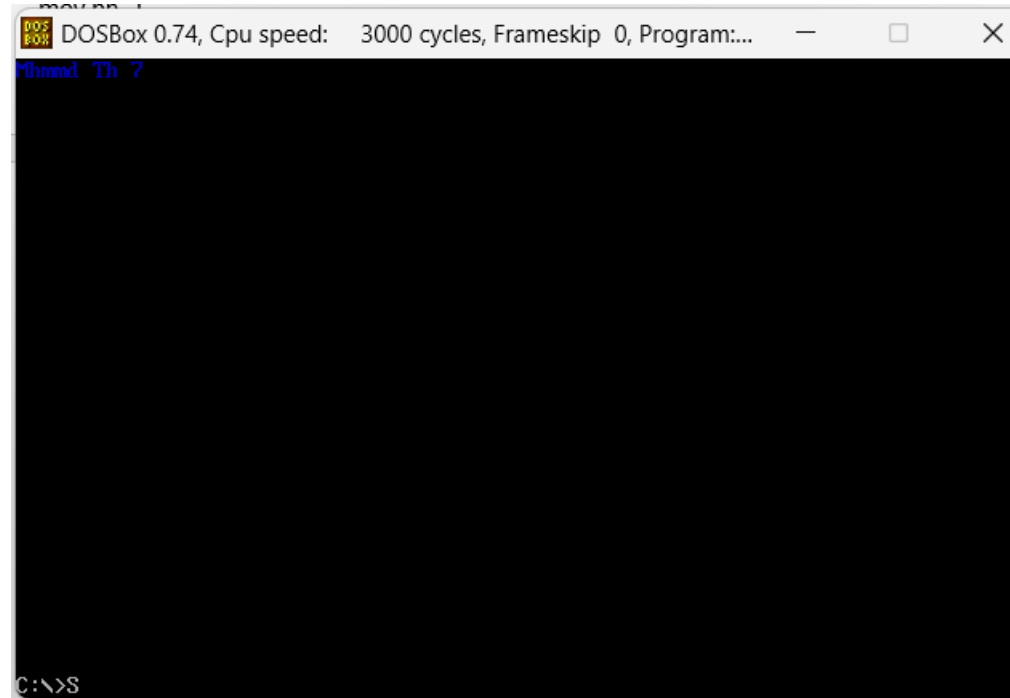
```
mov ax, 13
push ax
```

```
mov ax, message
push ax
```

```
call print
```

```
mov ax,0x4c00
```

```
int 21h
```



Best of luck